# Week 25: Row and column shadows

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

Each problem's key repeats the main answer from the print guide, week-25-facilitator.pdf, whose source is editable/facilitator-src/build_guide.py; the catalogs, forcing arguments and held hints are on the guide page named in the key's first line (printed page numbers, after the unnumbered overview). The overview repeats the guide's overview page. Change both together; check_guide_copy.py checks that every sentence here is in the guide.

# Mathematical overview

## The mathematical core

A picture is a binary matrix on fixed labeled rows and columns: each cell contains zero or one counter. Its margins are whole-row and whole-column sums. Both totals equal the number of counters, but margins need not determine the picture.

A **switch** replaces the two occupied opposite corners of a rectangle by its two empty opposite corners; all other cells stay fixed. The selected rows and columns need not be adjacent. Every affected row and column loses and gains one counter, so every margin is preserved.

**Uniqueness:** A binary picture is the only one with its margins if and only if it has no legal switch. Equivalently, the sets of occupied columns in its rows are pairwise nested. The guide gives the nested-row proof; finite enumeration alone is not its justification.

**Exact two-row distance:** For two pictures with the same margins, columns of sum 0 or 2 are forced. Among the s columns of sum 1, let S and T be the top-row occupied column sets of two pictures. Their equal size k is fixed by the row sum. The fewest switches is $|S \setminus T|$. One switch corrects at most one wrong top choice, and exchanging a wrong choice for a missing one attains the bound. All such pictures are connected; their greatest shortest distance is $\min(k, s-k)$.

## What the activity develops

**K–1:** Match line counts, find alternatives and organize small catalogs after a demonstrated picture-to-count conversion.

**Grades 2–3:** Discover ambiguity, follow a switch, and justify why a proposed route cannot be shorter.

**Grades 4–5:** Separate forced columns from free choices and prove the two-row distance rule for any number of columns.

## From trials to explanation

Two different matching pictures prove ambiguity. Failure to find a second picture does not prove uniqueness. A complete catalog, forcing argument or no-switch proof can do so. The distance formula above is for two rows; the guide cites broader switch connectivity separately.

# Problems and routes

| Problem | K–1 | 2–3 | 4–5 | route |
|---|---|---|---|---|
| Two different pictures for each pair of counts | P1 | | | first |
| Partner puzzles: make a picture, then find a different one | P2 | | | first |
| Every picture for rows 1, 2 and for rows 2, 1 | P3 | | | first |
| Every three-counter picture with one in each row and column | P4 | | | first |
| When can two different pictures share the counts? | P5 | | | first |
| Make counts your partner cannot fit with a different picture | P6 | | | return |
| Two matching pictures, if possible | | P1 | | first |
| Every picture with row and column counts 2, 1, 1 | | P2 | | first |
| Move two counters, keep the counts | | P3 | | first |
| Counts no partner can match with a different picture | | P4 | | first |
| Every switch in four pictures | | P5 | | first/return |
| Exactly one switch, exactly two switches | | P6 | | return |
| The six two-row pictures | | | P1 | first |
| Switch routes among the six pictures | | | P2 | first |
| A shortest route on six columns | | | P3 | first/return |
| Fixed columns and four free columns | | | P4 | first/return |
| The two pictures farthest apart on a two-row grid | | | P5 | return |
| The fewest switches on any two-row board | | | P6 | return |

**First route.** K–1 P1–P5; Grades 2–3 P1–P5; Grades 4–5 P1–P4. **Return visits.** K–1 P6; Grades 2–3 P5–P6; Grades 4–5 P3–P6.

**Page numbers.** A key that names a student page counts the pages of the full band packet. The route packets keep the problem numbers but number their own pages; each route-packet page is this page of the full packet:

| Route packet | its pages = full-packet pages |
|---|---|
| K–1 first | pp. 1–7 = pp. 1–7 |
| K–1 return | p. 1 = p. 1; p. 2 = p. 8 |
| Grades 2–3 first | pp. 1–7 = pp. 1–7 |
| Grades 2–3 return | p. 1 = p. 1; pp. 2–4 = pp. 6–8 |
| Grades 4–5 first | pp. 1–7 = pp. 1–7 |
| Grades 4–5 return | p. 1 = p. 1; pp. 2–6 = pp. 4–8 |

# Key, one entry per problem

## Two different pictures for each pair of counts

K–1 P1 · first route

*Idea.* The same row and column counts can fit different pictures: these three sets of counts fit 2, 2 and 3 pictures.

*Print guide, p. 3.* **Answer:** every pair can be filled differently. The three cases have 2, 2, and 3 solutions respectively. Any two of the displayed solutions solve each pair. Pictures are different when any fixed labeled cell changes, even if one is a reflection of another.

## Partner puzzles: make a picture, then find a different one

K–1 P2 · first route

*Idea.* A puzzle maker's counts may or may not fit a second picture; the partner's search tests it.

*Print guide, p. 7.* **P2:** there is no single required picture. Make any four-counter picture on each left grid, enter all counts, and let the partner test whether a different picture exists. Either answer can be correct. The partner first checks the counts: a finger along each row and down each column to its circle, before the maker draws.

## Every picture for rows 1, 2 and for rows 2, 1

K–1 P3 · first route

*Idea.* With one counter in one row and one in each column, placing that lone counter decides the picture: three pictures for each set of counts.

*Print guide, p. 4.* **Answer:** three pictures for the left column of grids and three for the right. They are two separate catalogs with different row counts.

For the left grids, the single top counter must be in one of three cells. Once it is placed, the bottom counters must occupy the other two columns, because each column needs exactly one. These are all three choices. The same argument with the bottom counter gives the right catalog.

## Every three-counter picture with one in each row and column

K–1 P4 · first route

*Idea.* One counter in each row and column of a 3-by-3 board gives 3 x 2 x 1 = 6 pictures: A chooses a column, B one of the two left, C the last.

*Print guide, p. 5.* **Answer:** exactly six. The first four recording grids are on page 5; the remaining two are at the top of page 6. The row letters stay fixed.

The A counter can occupy column 1, 2, or 3. For each choice, B can use either of the two remaining columns. C must use the last column. Thus there are two pictures for each of three choices: 2 + 2 + 2 = 6.

## When can two different pictures share the counts?

K–1 P5 · first route

*Idea.* A full or empty row or column can force every cell: two of these three sets of counts fit only one picture, the third fits five.

*Print guide, p. 6.* **Answers in printed order:** no, no, yes. The solution counts are 1, 1, and 5. A failure to find another picture is not yet a reason that none exists; use the forcing arguments below.

**Page 6:** row A needs all three cells. Column 1 needs two counters, so B1 is occupied; B2 and B3 must be empty. **Page 7 top:** row C and column 3 are empty. Column 1 needs two counters, forcing A1 and B1. B is now full, so the remaining counter in column 2 is A2. Each case forces every cell.

## Make counts your partner cannot fit with a different picture

K–1 P6 · return route

*Idea.* A picture is the only one with its counts exactly when no two counters can trade opposite corners of a rectangle; a full row forces cells.

*Print guide, p. 7.* **P6:** the maker tries for counts that fit only one picture; the partner tries to make a different picture with the same counts on the right grid.

**Pictures the partner cannot match:** A full, plus B1. The 2-by-3 counts are 3, 1 and 2, 1, 1; the 3-by-3 counts are 3, 1, 0 and 2, 1, 1. Column 1 then forces B1, and B has no other counter; on the 3-by-3 board, C is empty. Of the 15 four-counter pictures on the 2-by-3 grid, 9 fit only their own counts; on the 3-by-3 grid, 45 of 126 do. A partner's success

proves "more than one"; a partner's failure alone proves nothing, so check a claimed win with a forcing argument, or privately with the no-switch test on guide page 18.

## Two matching pictures, if possible

Grades 2–3 P1 · first route

*Idea.* Counts can fit several pictures or only one: these three sets fit 3, 1 and 2, and a full row or column forces cells.

*Print guide, p. 8.* **Answers from top to bottom:** yes, no, yes. The three margin sets have 3, 1, and 2 solutions.

Row A is full. The column count 2 forces B1; B has no counter left for another column. Therefore a different matching picture is impossible.

## Every picture with row and column counts 2, 1, 1

Grades 2–3 P2 · first route

*Idea.* Row A chooses two of three columns, and each choice leaves 1, 2 or 2 ways below: 1 + 2 + 2 = 5 pictures.

*Print guide, p. 9.* **Answer:** exactly five. Six recording grids are available; the unused grid is intentional space, not evidence of a sixth solution.

Row A occupies exactly two of three columns, so it has exactly three possible patterns. If A uses columns 2 and 3, the remaining column counts are 2, 0, 0; both B and C must use column 1. This gives one picture.

If A uses 1 and 3, the remaining counts are 1, 1, 0. B and C each need one counter, so they use columns 1 and 2 in either order. This gives two pictures. If A uses 1 and 2, the remaining counts are 1, 0, 1, giving two more in the same way. These cases are disjoint and exhaust row A, so there are 1 + 2 + 2 = 5.

## Move two counters, keep the counts

Grades 2–3 P3 · first route

*Idea.* Moving two counters to the empty corners of a rectangle keeps every row and column count: the first picture allows one such move, the second three.

*Print guide, p. 10.* **Page 3:** the only alternative moves A1 and C3 to A3 and C1. The selected rows and columns are nonadjacent. All of row B, column 2, and the other cells remain unchanged.

**Page 4:** the filled picture A1, B2, C3 has exactly three two-counter moves, one for each pair of rows: A1, B2 to A2, B1 (giving 010/100/001); A1, C3 to A3, C1 (001/010/100); B2, C3 to B3, C2 (100/001/010), with slashes separating rows A/B/C. Its three blank grids hold exactly these.

Moving two counters while keeping every count always uses the opposite corners of a rectangle, so a move needs two rows and the two columns their counters sit in. The other two pictures with counts 1, 1, 1 (010/001/100 and 001/100/010) move all three counters and are not answers.

## Counts no partner can match with a different picture

Grades 2–3 P4 · first route

*Idea.* A full row with the column counts below it forces every cell, so a picture can be the only one with its counts.

*Print guide, p. 10.* **Yes, a different matching picture can be made impossible on both shapes.** These four-counter choices are valid.

In each board A must be full. Column 1 requires B1. The 3-by-3 board has empty row C; all remaining cells are then forced empty. This establishes uniqueness. Other answers are welcome; check them using forcing or the criterion on guide page 18. A failed partner search alone proves nothing.

## Every switch in four pictures

Grades 2–3 P5 · first/return route

*Idea.* A switch needs a column occupied in one row and not the other, and one the other way round; pictures whose rows nest have none.

*Print guide, p. 11.* **Complete switch list:** children record a switch by drawing its rectangle through the four corner cells, as in the page 6 example. Top right has rows A,C and columns 1,3. The picture A1, A2, B3, C1 has rows A,B with columns 1,3; rows A,B with columns 2,3; and rows B,C with columns 1,3. Their results are respectively 011/100/100, 101/010/100, and 110/100/001, with slashes separating A/B/C. There are no others.

**Why the list is complete:** for each pair of rows, a switch needs a column occupied only in the first row and a column occupied only in the second. In the picture A1, A2, B3, C1, A versus B gives 2 choices; A versus C gives none because C is contained in A; B versus C gives 1. The top-left picture and the bottom staircase have nested occupied-column sets for every row pair, so no switch exists.

**Page 7** has four copies of the fourth picture, A1, A2, B3, C1, and no words of its own. Tell the children that these copies are extra workspace for further attempts: draw one switch's rectangle on each copy, and move on to a fresh copy when one gets crowded. The picture has three switches, so one copy stays spare.

## Exactly one switch, exactly two switches

Grades 2–3 P6 · return route

*Idea.* One switch replaces one top-row column, so a picture whose top row shares no column with the start needs two switches.

*Print guide, p. 12.* **One complete answer:** use the middle picture below on the middle grid and the last picture on the bottom grid. The route switches columns 1,3 and then 2,4.

**Why two cannot be one:** initially A occupies columns 1 and 2; in the last picture it occupies 3 and 4. One switch removes only one top-row counter from its old column and puts it in one new column. Two unwanted top positions must be removed, so at least two switches are necessary. The displayed route achieves two.

## The six two-row pictures

Grades 4–5 P1 · first route

*Idea.* With one counter in each column, the two top columns decide the picture: 3 + 2 + 1 = 6 pictures.

*Print guide, p. 13.* Eight recording grids are printed for the six pictures; the two spares are intentional space, not evidence of more pictures. Every column contains exactly one counter, so choosing the two occupied top columns determines the bottom row. The six possible pairs are 12, 13, 14, 23, 24, and 34. One way to prove completeness is to list pairs by their smaller element: three beginning with 1, two beginning with 2, and one beginning with 3. Total: 3 + 2 + 1 = 6.

## Switch routes among the six pictures

Grades 4–5 P2 · first route

*Idea.* A switch trades one top column for another, so the six pictures are connected and the greatest shortest distance is 2.

*Print guide, p. 13.* A legal switch exchanges one selected top column with one unselected top column. If two pictures are identical, distance is 0. If their top pairs share one column, switch the unwanted top column with the missing one; distance is 1. If the pairs are disjoint, replace one unwanted column, then the other; distance is 2.

The only distance-two pairs are {1,2} versus {3,4}, {1,3} versus {2,4}, and {1,4} versus {2,3}. Each needs at least two moves because one switch replaces just one top position. Every distinct pair fits one of these two cases, so all pictures

communicate and the largest shortest distance is exactly 2.

## A shortest route on six columns

Grades 4–5 P3 · first/return route

*Idea.* Three top counters sit where the target has none and a switch fixes at most one of them, so three switches are needed and enough.

*Print guide, p. 14.* **Answer:** exactly 3 switches. Student page 5 has three blank grids: a shortest route needs two for its intermediate pictures, and the third is spare. The route below switches column pairs 1,4; then 2,5; then 3,6. These are all legal two-row rectangles.

The target has no top counter in columns 1, 2, or 3. The start has one in each of those columns. One switch can remove a top counter from at most one of these three unwanted columns, so every route uses at least three switches. The displayed three-switch route meets this lower bound exactly.

## Fixed columns and four free columns

Grades 4–5 P4 · first/return route

*Idea.* A full and an empty column are forced, so only four free columns choose: 6 pictures, greatest shortest distance 2.

*Print guide, p. 15.* **Answers:** the printed endpoints are 2 switches apart; there are 6 pictures with these counts; the greatest shortest distance is 2. Column 1 is full and column 3 is empty in every solution. The only choices are columns 2, 4, 5, 6; exactly two must have their counter in row A. Page 7 has three blank grids with these counts as workspace, fewer than the six pictures, so the page does not give the count away; a child may list free top pairs instead of drawing every picture.

Start with free top pair {2,4}. Switch columns 2,5 to obtain {4,5}, then columns 4,6 to obtain {5,6}. The two unwanted top columns, 2 and 4, each have to be removed; a single switch removes at most one. Therefore two is both necessary and sufficient.

## The two pictures farthest apart on a two-row grid

Grades 4–5 P5 · return route

*Idea.* On a 2-by-6 grid one switch fixes at most one wrong top column, and at most three top columns can be wrong, so two pictures are never more than three switches apart.

*Print guide, p. 16.* **Answer:** 3 switches, for example 111000 / 000111 and 000111 / 111000 (top row / bottom row), the endpoints of Problem 3. A pair can set this as a challenge for each other, with the third child checking counts and switches.

**Why no pair needs more:** columns with count 0 or 2 are the same in both pictures. Among the s columns with count 1, a wrong top column is one occupied in the start's top row and empty in the target's. If the top row holds k of them, at most k can be wrong and at most s - k can be missing, so at most min(k, s - k) are wrong, which is at most 3 when s is at most 6. One switch fixes at most one wrong top column (Problem 3), and the route of the next problem fixes one with every switch.

## The fewest switches on any two-row board

Grades 4–5 P6 · return route

*Idea.* On two rows the fewest switches is the number of top counters in columns where the other picture has none: one switch fixes one, and pairing achieves it.

*Print guide, p. 17.* **Rule:** the fewest switches equals the number of columns with a top counter in the start and no top counter in the target. Count unwanted top positions, not all differing cells. If there are none, the pictures already agree and the answer is zero.

**3. Match one of each.** Choose an unwanted top column u and a missing top column v. Currently the counters are at Au and Bv, while Bu and Av are empty. Switching those corners puts both columns in their target states. It does not disturb any other column. Repeat until there are no unwanted positions. If there were d initially, this supplies a route of d switches.

**4. No route can do better.** Any two-row switch removes only one top counter and adds only one top counter, so it can reduce the number of unwanted top columns by at most 1. That number must fall from d to 0. At least d switches are required. The constructed route uses exactly d, proving the rule and connectivity for every number of columns.